

Spatial Omics Data Analysis

6-9 October, 2026

Jennifer Fransson

National Bioinformatics Infrastructure Sweden (NBIS)

'To help the Swedish research community build knowledge in analyzing large and complex omics data and to make bioinformatics easily accessible for life science researchers'



Support

Support services ranging from short consultations to long-term embedded bioinformaticians.



Infrastructure

Providing infrastructure in the form of services, computational resources, tools and guidelines to the life science community.



Training

Training events in advanced and applied bioinformatics.

ELIXIR

ELIXIR is an intergovernmental organisation that brings together life science resources from across Europe. These resources include databases, software tools, training materials, cloud storage and supercomputers.



ELIXIR Community: Single-Cell Omics

Community objectives

- Upskilling scientists in SCO data analysis and standards
- Developing benchmark proposals and collecting data suitable for benchmark creation
- Proposing best practices for FAIRification, establishing and promoting data and metadata standards

<https://elixir-europe.org/communities/single-cell-omics>

Trainers

Christophe Avenel (Scilifelab, SE)
Åsa Björklund (Scilifelab, SE)
Peiying Cai (UZH / SIB, CH)
Francesca Drummer (HZM, DE)
Jennifer Fransson (Scilifelab, SE)
Naveed Ishaque (BIH, DE)
Ahmed Mahfouz (LUMC, NL)
Benedetta Manzato (LUMC, NL)
Sergiu Netotea (Scilifelab, SE)
Lucie Pfeiferova (CAS, CZ)
Aditya Singh (Scilifelab, SE)
Suganya Sivagurunathan (Scilifelab, SE)
Leslie Solorzano (Scilifelab, SE)
Emma Vonk (LUMC, NL)
Jonas Windhager (Scilifelab, SE)

Organizers:

Ahmed Mahfouz (LUMC), ELIXIR-Netherlands
Eija Korpelainen (CSC), ELIXIR-Finland
Jennifer Fransson (Scilifelab), ELIXIR-Sweden
Christophe Avenel (Scilifelab, SE), ELIXIR-Sweden
Åsa Björklund (Scilifelab, SE), ELIXIR-Sweden

Schedule

10 Lectures
10 Practicals
1 Guest lecture

Topics covered

1. Overview of technologies
2. Cell segmentation
3. Preprocessing & QC
4. Segmentation-free analysis
5. Sequence-based analysis
6. Single cell data integration
7. Multi-slice integration
8. Spatial statistics
9. Domains and niches
10. Cell-cell communication

Coffee breaks

Take a break every morning and afternoon. If no break is indicated in the schedule, take a break during the practical exercises (coffee and snacks will be available)

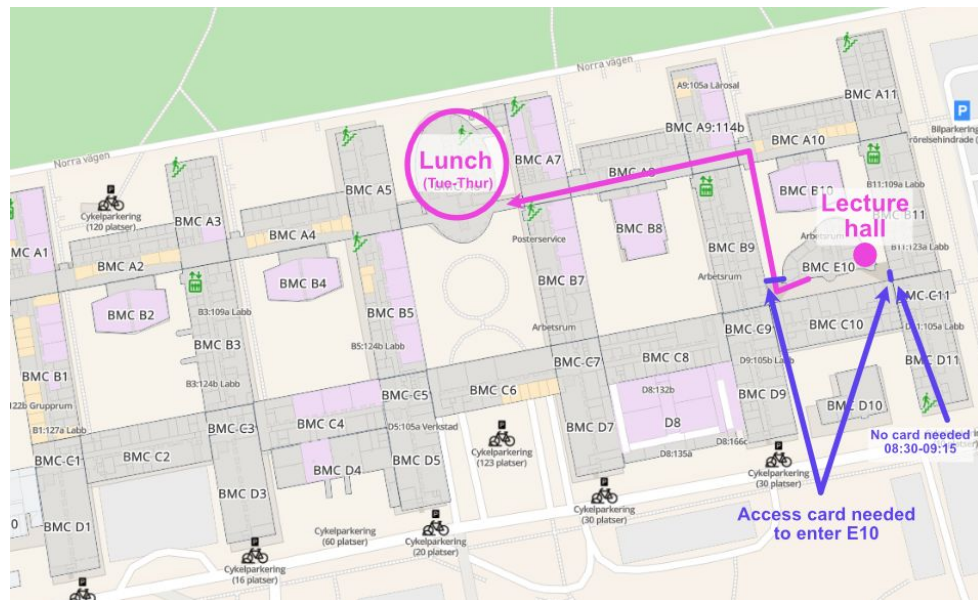
Logistics

- Lectures are recorded
- Attendance sheet
- Certificate of attendance
- Feedback form
- Photos

Lunch

At Bikupan (BMC's canteen)
Tuesday-Thursday

Access card is needed to
re-enter E10. Ask for help on
Slack (#help-me-im-lost) if you
can't enter



ELIXIR Code of Conduct

OUR VALUES

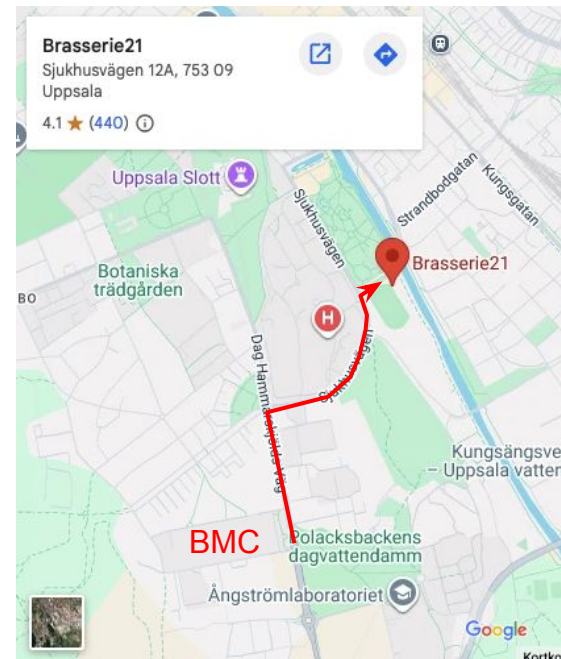
A place to feel respected A place to feel safe

The ELIXIR Code of Conduct has been created so all participants at ELIXIR events organised and /or funded by the ELIXIR Hub can interact with each other in a respectful and safe environment.

- We **value** each other's perspectives providing a safe environment for people to be themselves.
- We will **maintain** high ethical standards across all ELIXIR events.
- We **adopt** a zero-tolerance approach to harassment and discrimination in any form.
- We will **apply** honesty and integrity in the dealing of any transgressions against the Code.
- We are **committed** to making ELIXIR events a collaborative, supportive and enjoyable experience.
- We will **ensure** that our environment allows everyone to feel respected and included.

Course dinner tonight @ 17:30

- We will leave from BMC at 17:00
- Dietary requirements have been forwarded to the restaurant. Please ask the staff for help
- One beverage is included, order at the bar



Slack

- Logistics
- Questions about course content
- Issues
- Information sharing
- Social activities
- #help-me-im-lost

Serve accounts

Please let us know if you don't see this on your Serve page!

